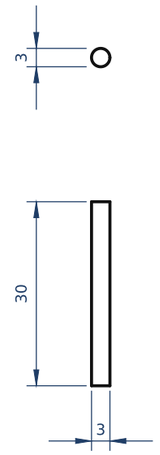
ENVELOPE (as the model draws it) 1:1



specification	value
rating	≥ 20 kV DC for the pump's nodes; ≥ 30 kV DC for the rings' leads (docs/rings-design.md §3)
candidate	FDH Aero UL3239-18-20KV-Z726 (tinned Cu 7/26, modified FEP, 20 kV, 200 °C)
rings' leads	a 30 kV UL3239 (listed by makers; part number to confirm) or a ≥ 40 kV DC silicone lead
model	drawn as Ø3 over a 1 mm conductor (the layout's)

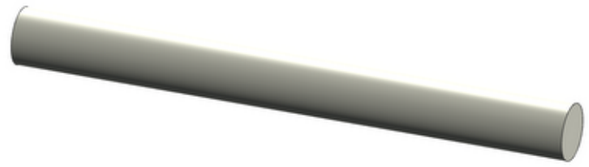
NOTES

1 UL3239's ratings are AC (a 15 kV rms spark test, 40 kV withstand): confirm the DC rating with the maker.

REVISIONS

Rev	Date	Description
A	2026-10-11	First issue, draft for quotation

status	BOUGHT
qty per machine	per DCK-501
used in	DCK-501
source	sim/multiplier_layout.py bore_wire (1 mm in 3 mm PTFE); docs/rings-design.md §3



3-D view (axonometric, not to scale)

DCCREG turbine · bought parts

HV wire, FEP, 20 kV, 18 AWG (UL3239)

Drawing no. **DCCREG-P-009**

Material: UL3239 high-voltage lead wire, FEP

Mass: —	Qty per machine: per DCK-501	Rev A Sheet 93 of 103 A3
Scale: 1:1	Tolerances: ISO 2768-mK	Date: 2026-10-11
Drawn: generated by docs/make_drawing_set.py	Checked: — Approved: —	**BOUGHT · SOURCE CONTROL**